MonkeyTail Six-Bar Pick-and-Place Mechanism

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Katie Stapleton: CAD, 3D printing, Aesthetic Design

Kamoliddin Sayfullaev: Linkage Synthesis, Kinematic Analysis, Actuator Selection

Description of Need

Automation is an important part of maintaining affordable prices for goods, maximizing the quantity of products companies can sell. However, in such processes, automating height gain can present an issue. A robotic arm is an unnecessary cost to simply raise the product from one platform to another. This is where the MonkeyTail Pick & Place mechanism comes into play.

The design of this pick and place mechanism allows for 90° movement. The item is transferred horizontally and vertically the same distance, and the toggle mechanism allows for continuous movement without the concern of reversing the motor direction. Though the linkage structure of the mechanism can have multiple applications, the job performed by this specific model would be to transfer small, lightweight objects. This could include items like figurines or bolts, though the intended application is pharmaceutical products like medication bottles. The inclusion of this product can help keep the costs of manufacturing down and thus keep the cost for the consumers down as well.

Project overview

The purpose of this project is to design a compact, repeatable pick-and-place mechanism that can move small parts from one location to another in a way that could realistically work next to a conveyor

belt in a light manufacturing/packaging setting. Our concept is based on a 90° pick-and-place motion, with continuous crank rotation driving the transfer and a separate servo operating the clamp.

The project progressed from two-position synthesis to CAD, kinematic and dynamic analysis, and a physical demo-prototype. The model predicts a peak input torque of 0.258 N·m at 20 rpm with a 100 g payload.

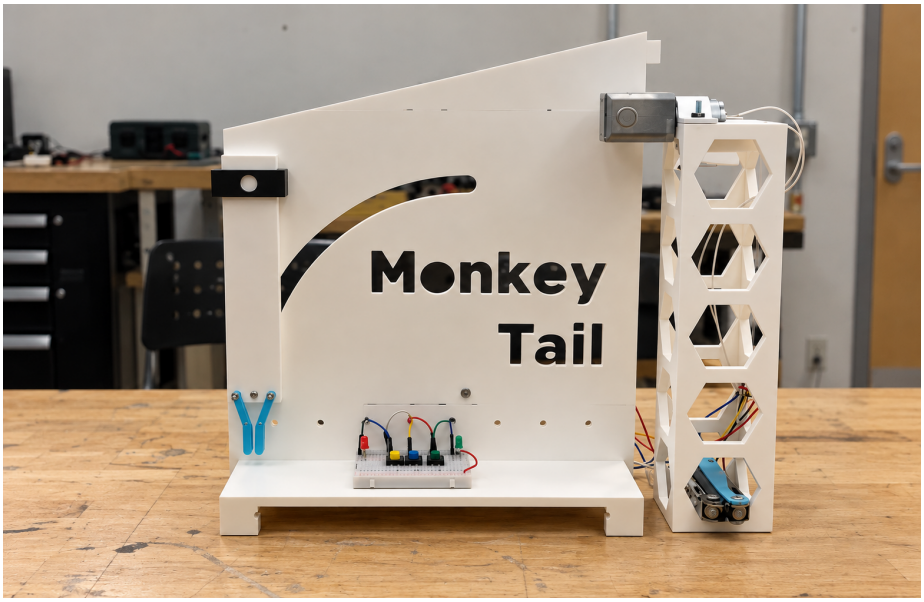


Figure 1. Demo prototype, front view.

Concept selection and CAD development

When we started the ideation process, we focused on how to achieve a smooth 90° pick-and-place motion without overcomplicating the design. We wanted something that is simple, repeatable, and realistic for a factory-style application. We chose a crank-rocker four-bar as the core of the motion because it provides constrained movement with one degree of freedom.

Two different linkages were synthesised for this problem: rocker and coupler outputs. Both were two-position problems. The rocker solution is a six-bar mechanism that involves a Grashof crank-rocker four-bar with a pinned slider that controls the direction of the gripper without separate orientation actuation. The coupler solution uses a non-Grashof four-bar driven by a second Grashof four-bar. The

rocker output was preferred for its simpler arrangement and adjustable task geometry. The clamp still requires its own servo.

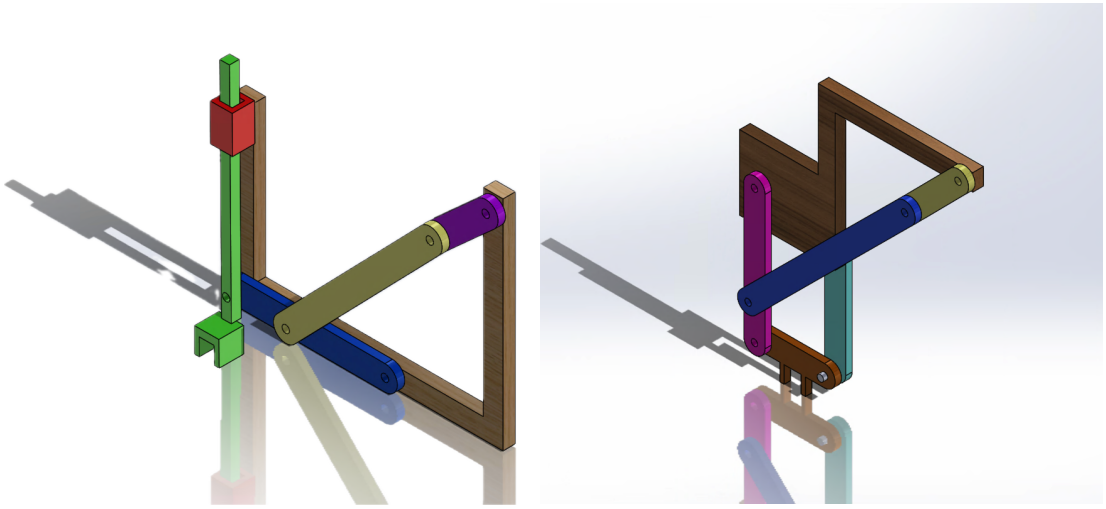


Figure 2. Rocker output concept (left) and coupler output concept (right).

To move closer to the aesthetic objectives, most linkages were relocated behind the frame. One smooth arc transfers the movement from behind the frame to the front. Platforms were added on either side, the top edge was angled, and the MonkeyTail name was incorporated into the frame. The CAD colors aid in organization; the physical model uses a white structure with a blue gripper.

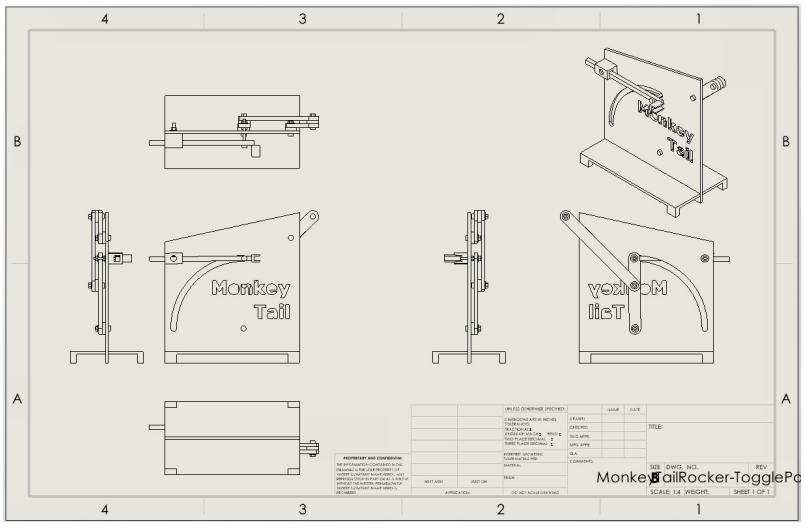


Figure 3. 7-View CAD drawing of the project

Geometry and analysis model

O4 is the origin, +x points right, +y points up, and positive rotation is counterclockwise. Fixed points are O2 ≈ (4.06, 7.76) in and O5 = (−6, 6) in. C is the rocker tip, E is the far end of the sliding bar, and P is the combined gripper and payload center of mass.

Member	Geometry in inches	Modeled mass
Ground O4–O2	8.76; orientation π/2.8851 rad	Fixed
Crank O2–A	Length 2.63; width 1	15.7 g
Coupler A–B	Length 8.34; width 1	42.2 g
Rocker O4–B–C	O4B = 3.70; O4C = 6; width 1	31.2 g
Sliding bar C–E	Length 7; width 1.5	48.7 g
Gripper and payload at P	2.5 × 2.5 × 1; P is 0.5 in beyond C	40 g + 100 g

Moving links are 6 mm thick. Link masses are from the original CAD volumes and acrylic density of 0.0433 lbm/in³. Centers of mass and inertias use explicit uniform rectangular-body estimates, including a combined bar, gripper, and payload body.

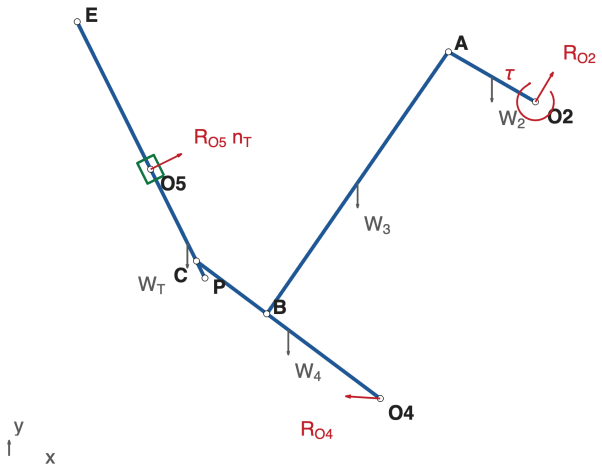


Figure 4. Free-body diagram of the six-bar mechanism at one sample configuration. Arrows indicate the force convention and are not drawn to scale.

The swivel is massless and frictionless. Its reaction is normal to the bar. The payload remains attached throughout the modeled cycle. Joint friction, fastening hardware mass, flexibility, and motor start/stop transients are omitted; these assumptions require checking on the prototype.

Kinematic results

Position is solved using circle intersections with a consistent assembly branch. Differentiated loop constraints provide velocity and acceleration. The bar orientation follows C–O5, and P is offset from C along that direction. This corrects the earlier simplified analysis that omitted the slider: its orientation affects the gripper center motion even though it does not change the rocker-tip path.

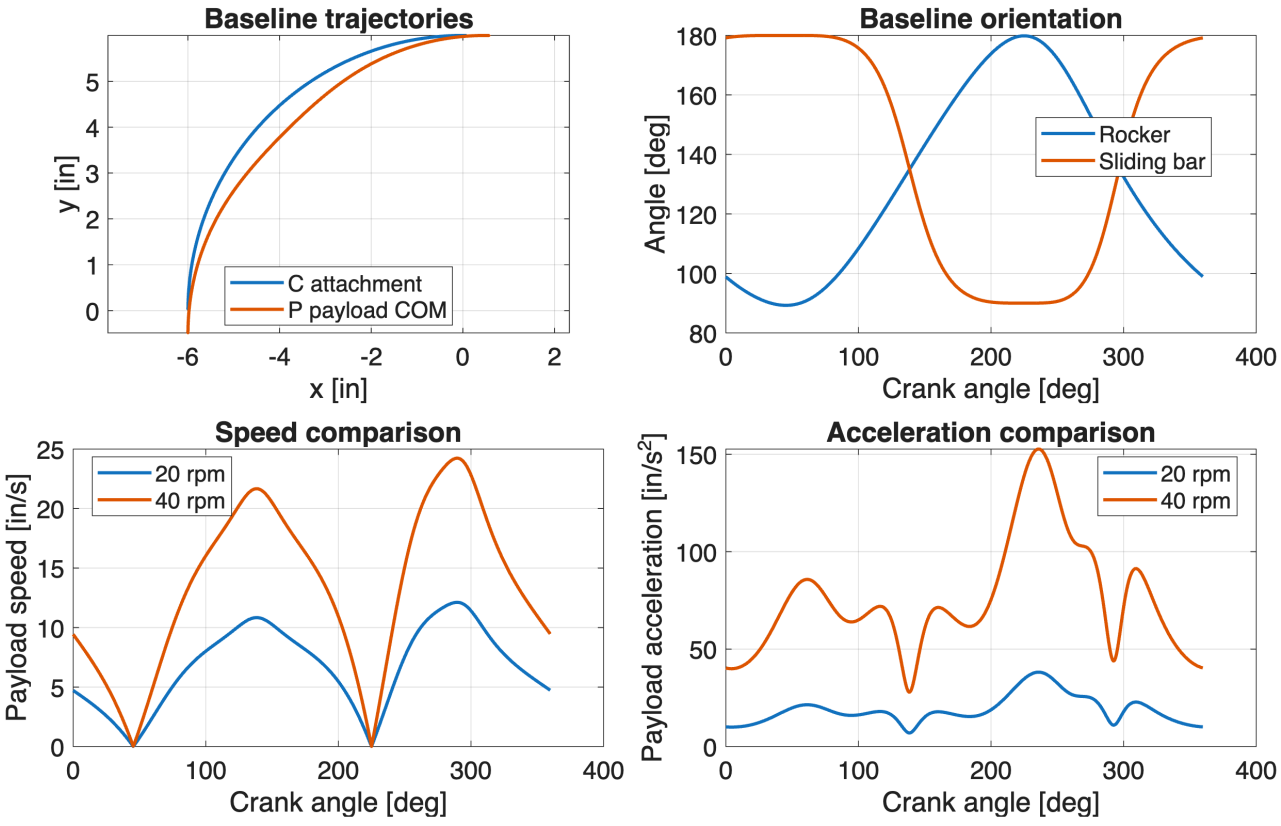


Figure 5. Trajectories, orientations, and gripper center speed and acceleration. Input angle is θ_2, measured from +x; C and P match Figure 4.

At the two rocker reversals	Horizontal separation	Vertical separation
Rocker tip C	6.08 in	5.99 in
Gripper and payload center P	6.58 in	6.49 in

The bar rotates approximately 90° between the reversals. Because the object center is offset from C, the modeled object transfer is not exactly 6 in. Pickup and drop-off fixtures must be aligned to the actual gripping point and verified physically. The C–O5 sliding distance ranges from 2.49 to 6.08 in, leaving at least 0.92 in of bar beyond the swivel in this ideal model.

Increasing input speed from 20 to 40 rpm doubles velocity and quadruples acceleration. Peak P speed rises from 12.12 to 24.23 in/s; peak acceleration rises from 38.17 to 152.68 in/s². The minimum four-bar transmission angle is 41.1°. The ratio ω_2/ω_4 diverges at rocker reversal because ω_4 is zero; this does not by itself mean the input-driven position solver is singular.

Dynamic forces and input torque

Each massive body is analyzed using $\Sigma F = m\, a_G$ and $\Sigma M_G = I_G\, \alpha$ about its center of mass. The crank, coupler, rocker, and combined terminal body supply twelve linear equations for the joint forces, ground reactions, swivel reaction, and input torque. The bar is treated as a translating and rotating body rather than rotating about a fixed material point at O5.

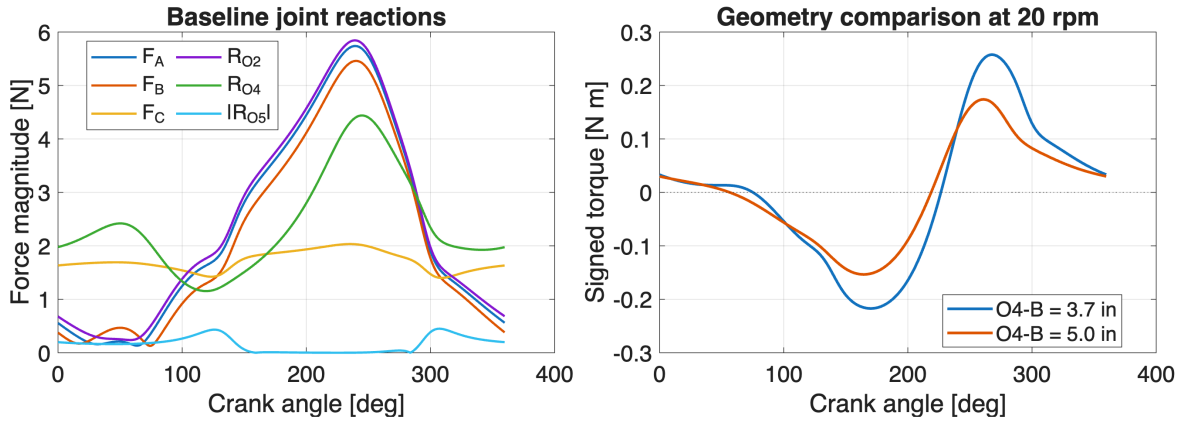


Figure 6. Joint-force magnitudes and signed input torque versus crank angle.

Force	Location	Peak N	Peak lbf	θ_2 at peak
F_A	A	5.738	1.290	239.5°
F_B	B	5.460	1.227	240.0°
F_C	C	2.033	0.457	235.3°
R_{O2}	O2	5.844	1.314	239.3°
R_{O4}	O4	4.439	0.998	244.8°
$\lvert R_{O5} \rvert$	O5	0.449	0.101	307.3°

At 20 rpm and 100 g payload, peak absolute input torque is 0.2576 N·m (2.280 lbf·in) at θ_2 = 267.75°; RMS torque is 0.1327 N·m. Negative torque represents portions of the cycle requiring braking or energy absorption. F_A acts on the crank, F_B on the coupler, and F_C on the rocker; the isolated diagrams in the appendix show the opposite reactions on adjacent bodies.

Design evaluation and actuator selection

Two design questions were investigated: how doubling the crank speed changes the motion, and how increasing O4B from 3.7 to 5.0 in changes the output while keeping O4C = 6 in fixed.

Configuration	Payload	Peak absolute torque
Baseline O4B = 3.7 in, 20 rpm	100 g	0.2576 N·m
Baseline O4B = 3.7 in, 40 rpm	100 g	0.2837 N·m
Baseline O4B = 3.7 in, 46 rpm	100 g	0.2974 N·m
Baseline O4B = 3.7 in, 20 rpm	1 lb	0.6879 N·m
Modified O4B = 5.0 in, 20 rpm	100 g	0.1739 N·m

Geometry decision

The longer O4B dimension reduces peak torque by about 32.5%, but it alters the pick and place locations. Between its rocker reversals, C moves only 4.72 in horizontally and 4.32 in vertically, and the bar rotates about 86.1°. Since reaching the intended task positions is the main purpose of the mechanism, the torque improvement was not worth losing the output motion. For this reason, the original O4B = 3.7 in design was kept for the final prototype.

Motor selection

The proposed actuator is the StepperOnline brushed 24 V DC gear motor CN-PA36-38244500-G71 [2]. Its listed rated output torque is 11 kgf·cm (1.079 N·m) at 46 rpm, including the 71:1 planetary gearbox. The rated-to-modeled peak torque ratio is approximately 4.19 at the 20-rpm baseline, or 3.63 for the 46-rpm simulation.

Prototype outcomes and conclusions

Our original plan was to make the base of the mechanism out of wood, but when we were unable to find the materials required, we turned to 3D printing the base. The printed base provided the aesthetic appeal we were looking for, but its functionality was lacking. The white linkages and base match our original goal of a smooth and cohesive mechanism, while the blue gripper makes its motion easier to observe.

Originally, we used traditional bolts and nuts, but the nuts continuously backed out. Nylon lock nuts were used to address loosening. The motor, motor driver, microcontroller, and wires remain visible, so improving the base and rewiring neatly are priorities for a future prototype.

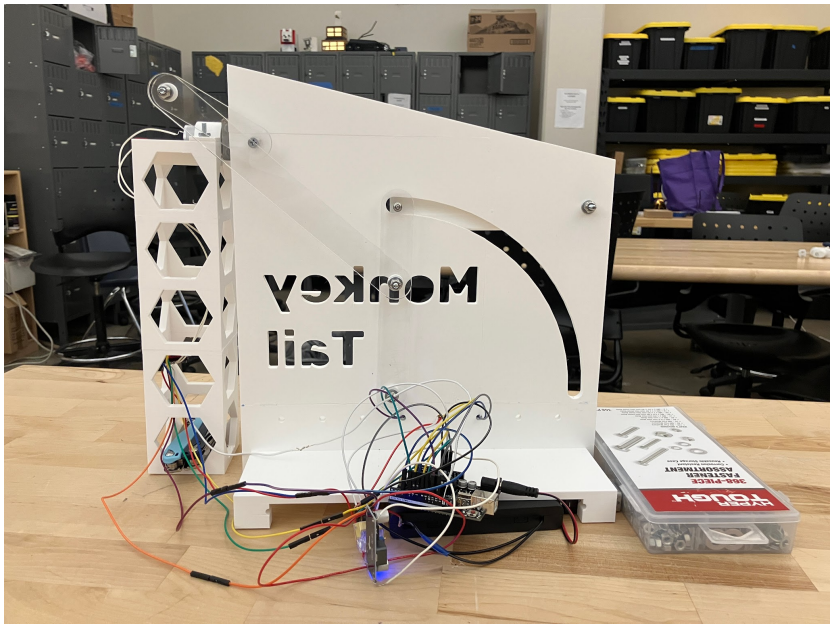


Figure 7. Demo prototype rear view, showing the drive, linkage arrangement, and exposed wiring.

A key lesson from this process was the importance of iteration and evaluation during the design stage. Small changes in link lengths or pivot locations can significantly affect motion. Starting each phase sooner and using a clearer timeline with milestones would allow more time for refinement and testing.

Conclusion

The MonkeyTail mechanism combines continuous crank rotation, a rocker output, and a swiveling slider to produce an approximately 90° transfer.

References

[1] thang010146. Pick and place mechanism 13. July 19, 2021. Concept inspiration.

[2] StepperOnline. CN-PA36-38244500-G71 24 V planetary gear motor specifications.

AI Disclosure

ChatGPT by OpenAI was used to convert hand-drawn Free Body Diagrams into clean 2D visuals.

Appendix Free body diagrams

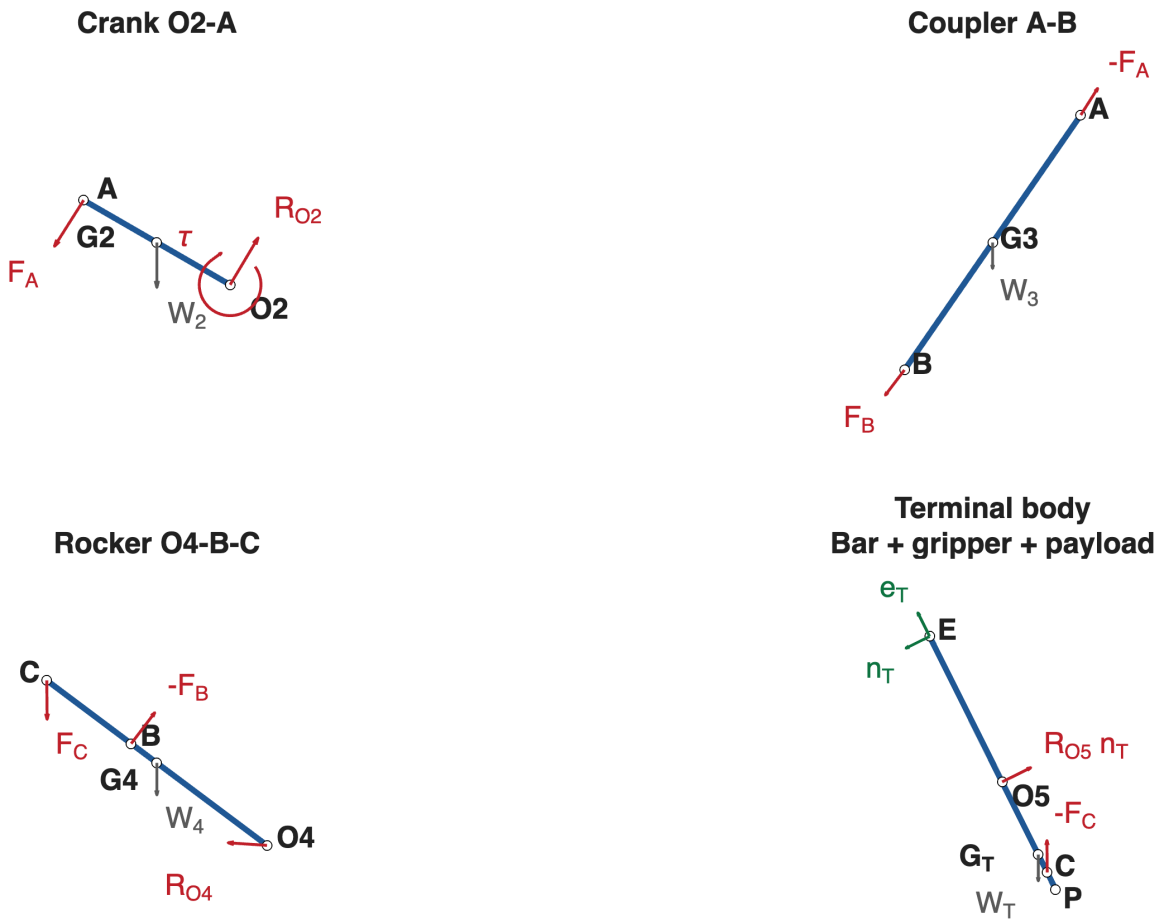


Figure 8. Isolated massive bodies using the same labels and force conventions as monkeytail_analysis.m. The swivel is the sixth link and is modeled as massless.

F_A acts on the crank at A; F_B acts on the coupler at B; F_C acts on the rocker at C. R_{O2} and R_{O4} are ground reactions. e_T points from C toward O5 and n_T is perpendicular to e_T; R_{O5} is the signed normal reaction. G2, G3, G4, and G_T are centers of mass, and $W_i = m_i\, g$. Arrow directions define positive forces and are not drawn to scale.